

3D Trigonometry — Problem Set

Note: Since this topic was previously in Year 11 Advanced Mathematics, all exam questions are sourced from Advanced Mathematics Examinations

Warm-up

1. A rectangular box has base dimensions 8 cm by 6 cm and a height of 5 cm. Find the length of the diagonal across the base.
2. A cube has side length s . Find the exact length of the internal diagonal connecting opposite vertices.
3. A vertical mast OT has height h . From point A on the ground, the angle of elevation to T is 25° . Express the distance OA in terms of h and a trigonometric ratio.
4. If $OA = h \cot 40^\circ$ and $OB = h \cot 50^\circ$, and $\angle AOB = 90^\circ$, write an expression for AB^2 using Pythagoras' Theorem.
5. In $\triangle PQR$, $PQ = 10$, $QR = 12$ and $\angle PQR = 60^\circ$. Use the Cosine Rule to find PR .
6. A point P is due North of O . A point Q is due East of O . State the size of $\angle POQ$.

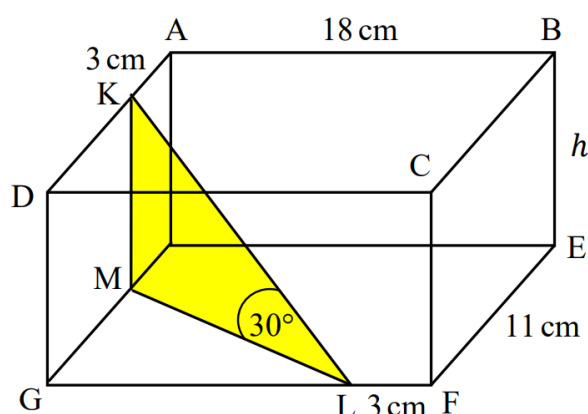
Skill-building

1. A tower OT of height h stands on level ground. From A , due West of O , the angle of elevation to T is 35° . From B , due South of O , the angle of elevation is 45° . Show that $AB^2 = h^2(\cot^2 35^\circ + 1)$.
2. In a rectangular prism with base $ABCD$ and height AE , $AB = 10$ and $BC = 15$. If the angle of elevation from C to E is α , show that $\tan \alpha = \frac{AE}{\sqrt{325}}$.
3. Two observers X and Y are 200 m apart on a North-South line. They both look at a balloon B directly above a point P . If P is due East of X , and the angles of elevation are 40° and 30° respectively, find h to 2 decimal places.
4. A flagpole h metres high stands at O . A is a point 50 m from O on a bearing of 040°T . B is a point 80 m from O on a bearing of 160°T . Find the distance AB and the angle of elevation from A to the top of the pole if $h = 20$.
5. A vertical mast PQ of height h is observed from A (bearing 000°T , elevation 20°) and B (bearing 090°T , elevation 30°). Find the bearing of B from A .
6. A hill is inclined at 15° to the horizontal. A path leads straight up the face of the hill. If a person walks 500 m up the path, what is their vertical rise?
7. From a point A , the bearing of a tower OT is 030°T . From B , 100 m due East of A , the bearing of the tower is 330°T . If the elevation from A is 25° , find the height h .
8. A pyramid has a rectangular base 12 cm by 8 cm. The edges from the apex to each corner are all 15 cm. Calculate the angle the slanted edge makes with the base.

9. Given $OA = h \tan 70^\circ$ and $OB = h \tan 60^\circ$, and $AB = 100$ where $\angle AOB = 45^\circ$, find h by applying the Cosine Rule to $\triangle AOB$.
10. Show that $h^2 \cot^2 \alpha + h^2 \cot^2 \beta - 2h^2 \cot \alpha \cot \beta \cos \theta = d^2$ is the general form for a tower of height h observed from two points distance d apart with elevation angles α, β and separated by angle θ at the base.

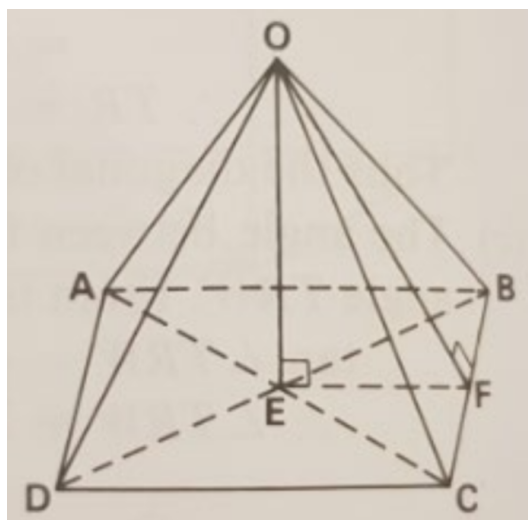
Easier Exam Questions

1. (St George Girls 2025 Trials Q9) The diagram shows the triangle KLM inside a rectangular prism. The dimensions are as shown.



What is the height, h , of the rectangular prism?

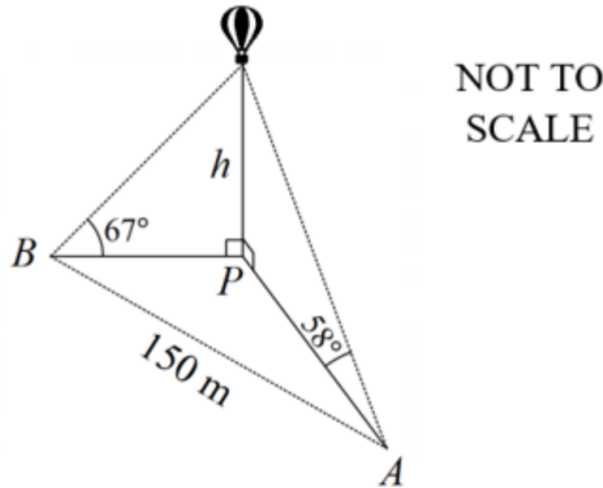
- (A) $17\sqrt{3}$ cm
 (B) $\frac{17\sqrt{3}}{3}$ cm
 (C) $15\sqrt{3}$ cm
 (D) $\frac{17\sqrt{3}}{2}$ cm
2. (Girraween 2020 Preliminary Q12c) The square pyramid shown below has a height of 16 cm and a base edge of 24 cm. Find:



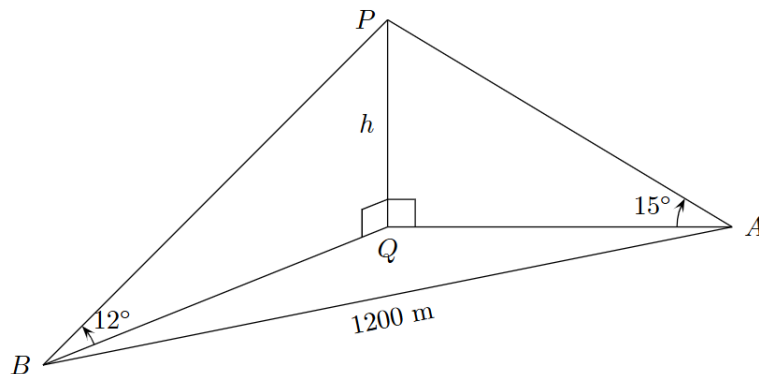
- (i) the length of OF .

- (ii) the size of the angle between the plane OBC and the base $ABCD$. 1
- (iii) the length DE and hence find the size of the angle between the edge OD and the base $ABCD$. 3

3. (Girraween 2022 Preliminary Q10b) A newly launched weather balloon is floating in the air vertically h metres above the point P on the ground. Observer A is due south of the balloon and notes that its angle of elevation is 58° . Observer B is 150 metres from Observer A and due east of the weather balloon. The angle of elevation of the weather balloon from Observer B is 67° .



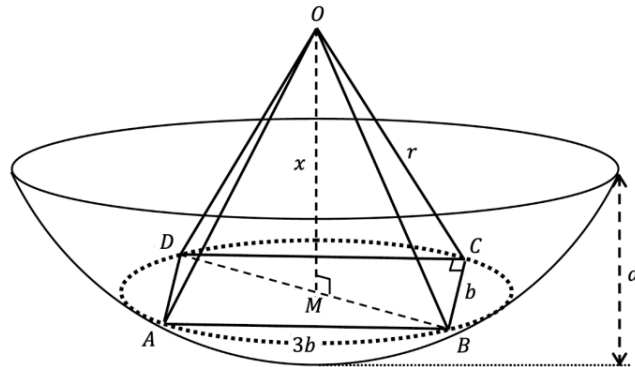
- (i) Show that $PA = h \cot 58^\circ$ and $PB = h \cot 67^\circ$. 2
- (ii) Use Pythagoras' Theorem to show that $h^2 = \frac{150^2}{\cot^2 58^\circ + \cot^2 67^\circ}$ 2
- (iii) Hence, find h correct to one decimal place. 1
4. (Normanhurst Boys 2023 Y11 Half-Yearly Q9) A tower PQ of height h metres has an angle of elevation of 15° from a point A due east of it. From another point B , the bearing of the tower is $055^\circ T$ and the angle of elevation is 12° . The points A and B are 1200 metres apart on the same level as the tower's base Q .



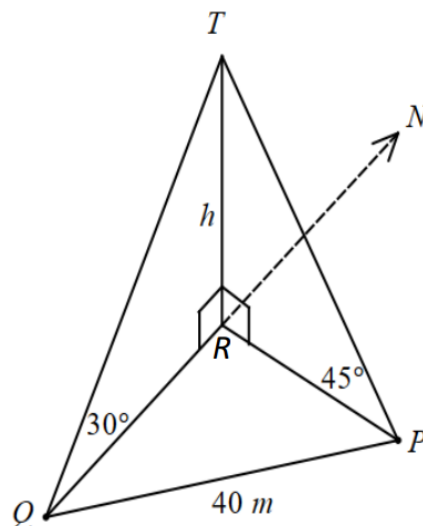
- (a) Show that $\angle AQB = 145^\circ$ 1
- (b) Consider the triangle APQ and show that $AQ = h \cot 15^\circ$. 1
- (c) Find a similar expression for BQ . 1

Harder Exam Questions

1. (Blacktown Boys 2024 Trials Q33a) A rectangular pyramid is placed in a bowl of depth d which is part of a sphere with centre O and radius r . The apex of the pyramid coincides with O . The vertices of the base, $ABCD$, lie on the bowl's inner surface as shown in the diagram. The perpendicular from O meets the diagonal DB in its midpoint M . Let $OM = x$, $CB = b$, and $AB = 3b$.

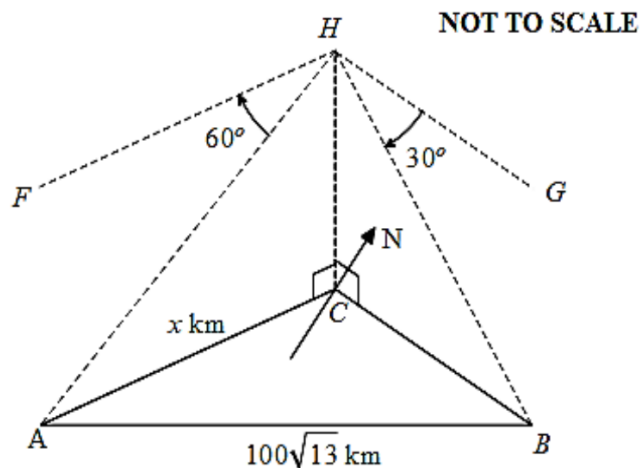


- (a) Show that $DB = 2\sqrt{r^2 - x^2}$ 1
2. (Girraween 2025 Trials Q29) A vertical tower of height h metres stands on horizontal ground. From a point P on the ground due east of the tower the angle of elevation of the top of the tower is 45° . From a point Q on the ground due south of the tower the angle of elevation of the top of the tower is 30° . If the distance PQ is 40 metres, find the exact height of the tower. 4

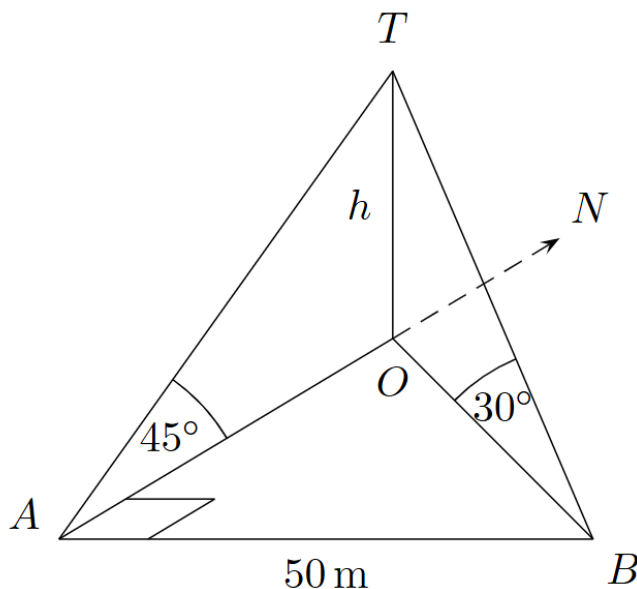


3. (Hornsby Girls 2025 Trials Q33a) The figure below shows three buoys at sea at A , B and C . A helicopter flies along FH and then HG , where F, G and H are three points of the same height vertically above A, B and C , respectively. When the helicopter reaches H , the angles of depression of A and B from the helicopter are 60° and 30° , respectively. The true bearings of A and B from C are 250° and 130° , respectively. The distance between A and B is $100\sqrt{13}$ km.

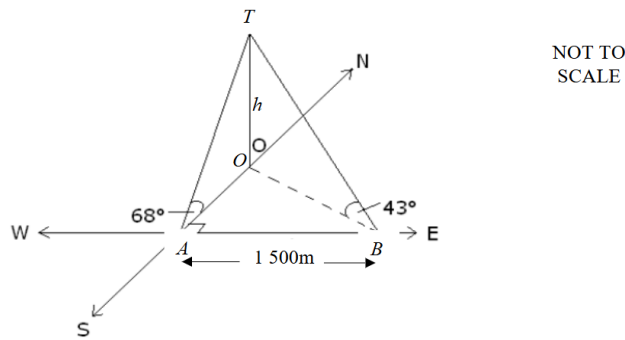
- (a) Show that the distance between A and C is 100 km. 3



4. (Normanhurst Boys 2024 Y11 Half-Yearly Q16) A person is standing at point A , which is due south of a tower OT that is h metres tall. The person walks 50 metres east to point B . The angle of elevation at points A and B to the top of the tower are 45° and 30° respectively.

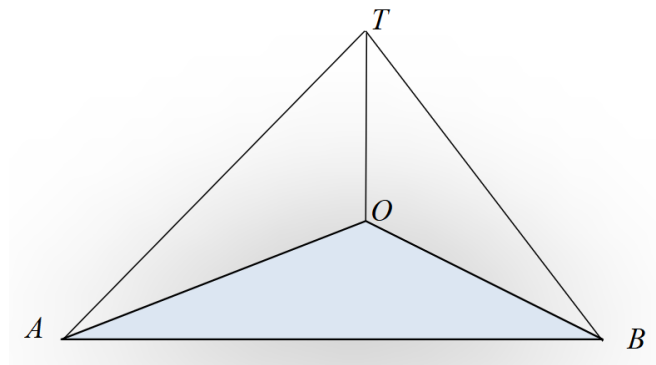


- (a) Express OA and OB in terms of h . 2
 - (b) Show that $h = 25\sqrt{2}$ metres. 2
 - (c) Find the bearing of B from the base of the tower O , correct to the nearest minute. 2
5. (Hornsby Girls 2021 Y11 Half-Yearly Q11b) A lighthouse of height h metres, is observed from two points, A and B , in the same horizontal plane as the base of the lighthouse. From point A , due south of the lighthouse, the angle of elevation to the top of the lighthouse is 68° . From point B , due east of point A , the angle of elevation to the top of the lighthouse is 43° . Points A and B are 1 500 metres apart.
- (i) Show that $OA = h \tan 22^\circ$ 2
 - (ii) Show that $h = \frac{1\,500}{\sqrt{\tan^2 47^\circ - \tan^2 22^\circ}}$ 3
 - (iii) Find the height of the tower correct to the nearest metre. 1



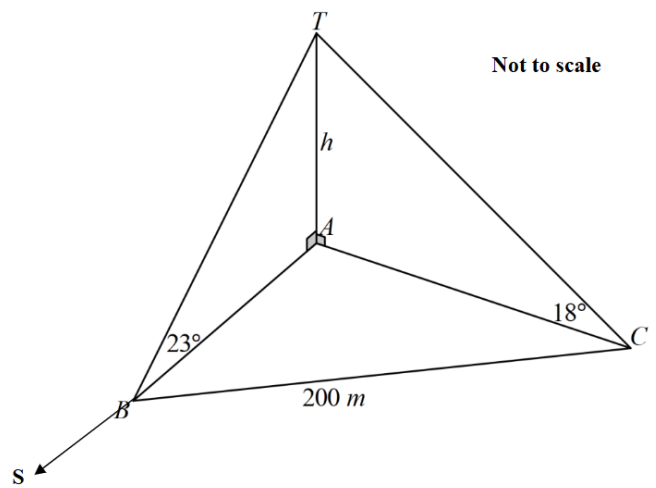
6. (Hornsby Girls 2023 Y11 Half-Yearly Q11b)

Alex walks along a straight road on level ground. At one point, A , a tower is noticed on a bearing of 053° , with an angle of elevation of 21° . After Alex walks a further 230 m to a point B , the tower is on a bearing of 342° with an angle of elevation of 26° . Let the base of the tower be the point O and the top of the tower be the point T .

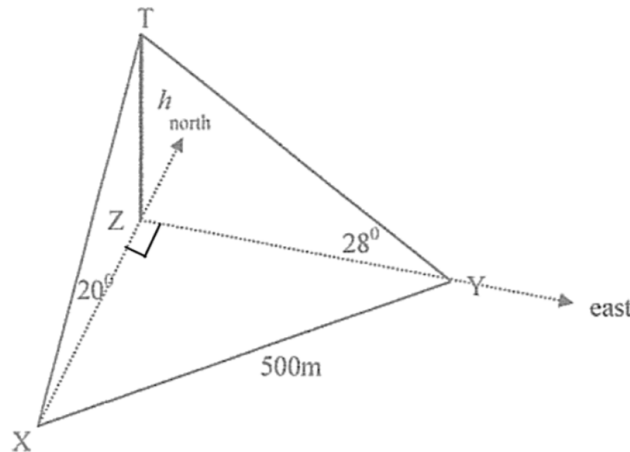


- (i) State the size of $\angle AOB$. 1
- (ii) Find the height of the tower correct to the nearest metre. 3

7. (Hornsby Girls 2024 Y11 Half-Yearly Q9d) TA is a vertical tower of height h metres. B and C are points at the same level as A , the foot of the tower, and $BC = 200$ m. The point B is due south of the tower and C is on a bearing of $120^\circ T$ from the base of the tower. The angles of elevation to the top of the tower from B and C are 23° and 18° , respectively. Find the height of the tower, correct to one decimal place. 4

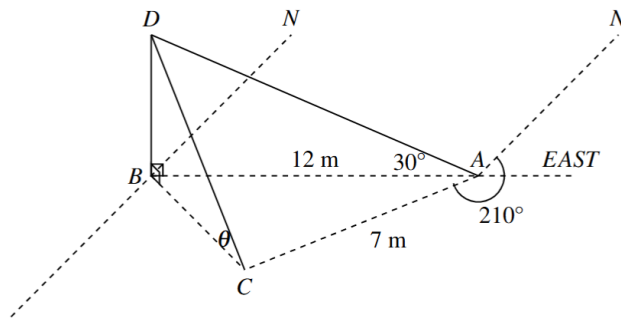


8. (Girraween 2023 Preliminary Q12b) A conservationist observes the angle of elevation of the top of a tree, which is h metres tall, from two positions. From a point X , due south of the tree, it is 20° and from point Y , due east of the tree, it is 28° . The distance XY is 500 m.



- (i) Write expressions for XZ and YZ in terms of h . 2
 - (ii) Calculate the value of h . 2
9. (Penrith 2023 Preliminary Q30)

A flagpole, BD , sits on flat level ground. It is supported by two cables, AD and CD , as shown below. The cable AD is inclined at 30° to the horizontal and is tied to the floor 12 metres from the foot of the flagpole. The second cable is tied to the floor at point C , 7 metres from A and on a bearing of 210° from A .



- (a) Show that $\angle BAC = 60^\circ$. 1
- (b) Find the exact height of the flagpole, BD . 1
- (c) Find the exact distance B to C and hence, find θ , correct to the nearest degree 3